CST2102 – Final Project

Hospital Database System

Group 7 | Algonquin College

1. Project Overview

This project designs and implements a relational database system for a hospital. It covers all aspects of hospital operations including patient management, doctor scheduling, medical records, prescriptions, billing, room assignments, and staff management.

The database is built in SQL Server and includes: DDL for all 11 tables with full constraints, synthetic DML data, 15 business queries with multi-table joins, and 6 database users with appropriate permissions.

2. Database Schema

2.1 Tables Overview

The database consists of 11 tables:

Table	Primary Key	Key Attributes
Department	INT PK	DepartmentName, Location, PhoneExtension
Patient	INT PK	FirstName, LastName, DOB, Gender, Address, Phone, Email, EmergencyContact*
Doctor	INT PK	FirstName, LastName, Specialization, Phone, Email, DepartmentID FK, Availability
Medicine	INT PK	MedicineName, Manufacturer, StockQuantity, Price
Staff	INT PK	FirstName, LastName, Role, DepartmentID FK, Phone, Email, ShiftHours
Room	INT PK	RoomNumber, DepartmentID FK, RoomType, AvailabilityStatus
Appointment	INT PK	PatientID FK, DoctorID FK, DepartmentID FK, Date, Time, Status
MedicalRecord	INT PK	PatientID FK, DoctorID FK, VisitDate, Diagnosis, TreatmentPlan, Prescription

Prescription	INT PK	RecordID FK, MedicineID FK, Dosage, Frequency, Duration
Billing	INT PK	PatientID FK, TotalAmount, PaymentStatus, PaymentDate, PaymentMethod
RoomAssignment	INT PK	RoomID FK, PatientID FK, AdmissionDate, DischargeDate

2.2 Relationships

- **One-to-Many:** Department → Doctor, Staff, Room, Appointment
- **One-to-Many:** Patient → Appointment, MedicalRecord, Billing, RoomAssignment
- **One-to-Many:** Doctor → Appointment, MedicalRecord
- **One-to-Many:** MedicalRecord → Prescription
- **Many-to-Many (via bridge):** Patients ↔ Doctors via Appointment; Medicines ↔ MedicalRecords via Prescription

2.3 Key Constraints

- **Primary Keys:** All tables use INT identity primary keys
- **Foreign Keys:** All FK relationships enforced with FOREIGN KEY constraints
- **Unique Constraints:** Email, PhoneNumber, RoomNumber, DepartmentName, MedicineName
- **NOT NULL:** All essential fields (FirstName, LastName, Dates, Status fields)
- **CHECK Constraints:** RoomType, AvailabilityStatus, PaymentStatus, Appointment Status, Price > 0, Stock >= 0, DischargeDate >= AdmissionDate

3. Synthetic Data (DML)

All data was programmatically generated using Python with realistic names, addresses, dates, and medical terminology. Record counts per table:

Table	Records	Notes
Department	8	All 8 hospital departments
Patient	200	Synthetic patients with demographics
Doctor	40	5 per department, varied specializations
Medicine	80	Real medicine names with pricing
Staff	75	Nurses, admin, technicians, etc.
Room	40	Mix of General, Private, ICU, OR, Recovery
Appointment	800	70% Completed, 20% Scheduled, 10% Cancelled
MedicalRecord	700	Linked to patients and doctors
Prescription	1500	Multiple prescriptions per medical record
Billing	800	70% Paid, 30% Unpaid

RoomAssignment	300	Patient room stays with admission/discharge dates

4. Business Questions & Queries

The following 15 business questions were formulated to demonstrate real hospital analytics. All queries use JOINs across multiple tables.

#	Category	Business Question
Q1	Patient & Appointment Analysis	Average appointments per patient in the last 6 months
Q2	Doctor Workload	Which doctors have the highest patient load vs availability?
Q3	Financial Overview	Total revenue by department in the last quarter
Q4	Prescription Patterns	Top 5 most prescribed medicines in the last 3 months
Q5	Room Utilization	Occupancy percentage by room type in the last month
Q6	Patient Admissions	Patients admitted to a room more than once
Q7	Billing / Collections	List of all outstanding unpaid bills with patient contact info
Q8	Clinical Insights	Most common diagnoses across all medical records
Q9	Scheduling Trends	Appointments per department per month in 2024
Q10	Doctor Activity	Doctors with no appointments in the last 90 days
Q11	Payment Analysis	Average bill amount and total revenue by payment method
Q12	HR / Staffing	Staff count by role and department
Q13	Hospitalized Patients	Patients with both appointments and room admissions
Q14	Inventory Management	Medicine stock value and low-stock alerts
Q15	Full Patient Summary	Complete patient summary: appointments, records, billing, admissions

5. Database Users

One administrator and five read-only users were created. Read-only users are assigned to the db_datareader role which grants SELECT access only — they cannot INSERT, UPDATE, DELETE, or create/drop objects.

Username	Role	Permissions
hospital_admin	db_owner	Admin – full read/write/create/drop access
hospital_user01	db_datareader	Read-only – SELECT only
hospital_user02	db_datareader	Read-only – SELECT only

hospital_user03	db_datareader	Read-only – SELECT only
hospital_user04	db_datareader	Read-only – SELECT only
hospital_user05	db_datareader	Read-only – SELECT only

6. Assumptions & Notes

- **Billing is patient-level** (not per appointment) to simplify the billing model.
- **Room assignments** may overlap for the same room in the synthetic data; a production system would add a constraint to prevent this.
- **Prescription column in MedicalRecord** is a note field (e.g. "See prescription #N"); actual medicine details live in the Prescription table.
- **All data is synthetic** and generated for educational purposes only. No real patient information was used.
- **PowerBI dashboard** connects to the live SQL Server using the hospital_admin login and visualizes appointments, billing, room utilization, and prescription trends.

7. Potential Improvements

- **Add an Insurance table** to track patient insurance providers and reduce billing complexity.
- **Add a LabResult table** linked to MedicalRecord to store test results separately from diagnosis notes.
- **Add a RoomAssignment uniqueness constraint** to prevent double-booking the same room on overlapping dates.
- **Stored procedures** for common operations (admit patient, discharge patient, generate invoice) would improve maintainability.
- **Audit logging** via triggers on Billing and MedicalRecord would support compliance requirements.